

SKILL - 4

```
MySQL 8.0 Command Line Cli  ×  +  v

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.22 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE skill4_db;
Query OK, 1 row affected (0.02 sec)

mysql> USE skill4_db;
Database changed
mysql> SHOW TABLES;
Empty set (0.00 sec)

mysql> CREATE TABLE accounts (
->     account_id INT PRIMARY KEY,
->     customer_name VARCHAR(50),
->     balance DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO accounts VALUES
-> (1, 'Ravi', 5000.00),
-> (2, 'Sita', 3000.00),
-> (3, 'Kiran', 7000.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```



```
mysql> SELECT * FROM accounts;
```

account_id	customer_name	balance
1	Ravi	5000.00
2	Sita	3000.00
3	Kiran	7000.00

```
3 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE accounts
```

```
-> SET balance = balance - 1000
```

```
-> WHERE account_id = 1;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE accounts
```

```
-> SET balance = balance + 1000
```

```
-> WHERE account_id = 2;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT * FROM accounts;
```

account_id	customer_name	balance
1	Ravi	4000.00
2	Sita	4000.00
3	Kiran	7000.00

```
3 rows in set (0.00 sec)
```

```
mysql> SAVEPOINT transfer_done;
```

```
Query OK, 0 rows affected (0.00 sec)
```



```
mysql> UPDATE accounts
      -> SET balance = balance - 500
      -> WHERE account_id = 3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_id | customer_name | balance |
+-----+-----+-----+
|          1 | Ravi          | 4000.00 |
|          2 | Sita          | 4000.00 |
|          3 | Kiran         | 6500.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> ROLLBACK TO SAVEPOINT transfer_done;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_id | customer_name | balance |
+-----+-----+-----+
|          1 | Ravi          | 4000.00 |
|          2 | Sita          | 4000.00 |
|          3 | Kiran         | 7000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> COMMIT;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM accounts;
```

account_id	customer_name	balance
1	Ravi	4000.00
2	Sita	4000.00
3	Kiran	7000.00

```
3 rows in set (0.00 sec)
```

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE GetAccountBalance(IN acc_id INT)
```

```
    -> BEGIN
```

```
        -> SELECT account_id, customer_name, balance
```

```
        -> FROM accounts
```

```
        -> WHERE account_id = acc_id;
```

```
    -> END //
```

```
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> CALL GetAccountBalance(1);
```

account_id	customer_name	balance
1	Ravi	4000.00

```
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION GetBalance(acc_id INT)
```

```
    -> RETURNS DECIMAL(10,2)
```

```
    -> DETERMINISTIC
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetBalance(acc_id INT)
  -> RETURNS DECIMAL(10,2)
  -> DETERMINISTIC
  -> BEGIN
  ->     DECLARE bal DECIMAL(10,2);
  ->
  ->     SELECT balance INTO bal
  ->     FROM accounts
  ->     WHERE account_id = acc_id;
  ->
  ->     RETURN bal;
  -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT GetBalance(1) AS Ravi_Balance;
+-----+
| Ravi_Balance |
+-----+
|      4000.00 |
+-----+
1 row in set (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER check_balance
  -> BEFORE UPDATE ON accounts
  -> FOR EACH ROW
  -> BEGIN
  ->     IF NEW.balance < 0 THEN
  ->         SIGNAL SQLSTATE '45000'
  ->         SET MESSAGE_TEXT = 'Balance cannot be negative';
  ->     END IF;
  -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> UPDATE accounts
    -> SET balance = balance + 100
    -> WHERE account_id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM accounts;
```

account_id	customer_name	balance
1	Ravi	4100.00
2	Sita	4000.00
3	Kiran	7000.00

```
3 rows in set (0.00 sec)
```

```
mysql> |
```